

MS24A EPD

Electronic Personal Dosimeter

User Manual

Revision 1.2.5 | April 2026



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Product Improvements

Continuous improvement of products is a policy of CBRN PROTECTION TCT BV. All specifications and operating instructions are subject to change.

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1. Introduction and Scope

1.1 About This Manual

This user manual provides comprehensive instructions for the safe and effective operation, maintenance, and storage of the MS24A Electronic Personal Dosimeter (EPD) manufactured by CBRN PROTECTION TCT BV. It contains essential technical specifications, operating procedures, alarm descriptions, calibration guidance, and care instructions.

Read this manual thoroughly before using the dosimeter. Keep it accessible for future reference. This manual applies to all models of the MS24A EPD.

1.2 Product Overview

The MS24A is a direct-reading electronic personal dosimeter designed for measurement of personal dose equivalent $H_p(10)$ and personal dose equivalent rate $H^*_p(10)$ (hereinafter referred as DE and DER) from photon radiation, in accordance with ANSI N42.20 and IEC 61526:2024.

The dosimeter employs a Pb-doped polystyrene scintillator detector with energy compensation, providing tissue-equivalent energy response across a wide photon energy range. This technology offers sensitivity comparable to large-volume GM tubes while maintaining the compact form factor required for personal dosimetry.

1.3 Intended Use

The MS24A is intended for use as an electronic personal dosimeter (EPD) for occupational radiation monitoring. It may be used as a standalone device or integrated within automated individual dose monitoring systems. Typical applications include:

- Nuclear power plant personnel and related nuclear industry workers
- Emergency response teams, firefighters, and law enforcement
- Medical personnel in radiology, nuclear medicine, and radiation therapy
- Research reactor and laboratory personnel
- Personnel handling radioactive materials in processing and transportation
- Security checkpoint operators at facilities using X-ray or accelerator-based scanners

1.4 Key Features

- Tissue-equivalent Pb-doped plastic scintillator detector with energy-compensated response
- Wide dose equivalent rate measurement range: 0.1 $\mu\text{Sv/h}$ to 10 Sv/h
- Wide dose equivalent measurement range: 1 μSv to 20 Sv
- Audible, visual (LED), and vibration alarm indicators for dose and dose rate thresholds
- Non-volatile memory storage for up to 2,000 events

- USB and RS-485 data interfaces
- Extended battery life: minimum 3,000 hours (approximately 90 days continuous operation) from a single AA battery
- Compact and lightweight: 23 × 63 × 91 mm, 85 g (without clip and battery)
- IP67 ingress protection rating per IEC 60529
- Drop-resistant: withstands 1.5 m drop onto concrete
- Designed for group charging and readout at NPPs

2. Safety Information

2.1 General Safety Precautions

Observe the following precautions at all times when handling or operating the MS24A dosimeter:

- All operations involving calibration, verification, repair, or maintenance using radioactive sources must be performed in compliance with applicable radiation safety regulations and basic health physics practices.
- Do not expose the dosimeter to corrosive or chemically aggressive environments.
- Do not expose the dosimeter to temperatures exceeding the rated operating range (-20°C to $+50^{\circ}\text{C}$).
- Avoid mechanical shock, impacts, and physical damage beyond the rated 1.5 m drop height.
- Do not expose the dosimeter to intense electromagnetic interference beyond its rated EMC immunity (Group IV per IEC 61000-6-2).
- Although rated IP67, avoid prolonged or intentional submersion in water or other liquids.

2.2 Battery Safety

- Use only AA-size (FR6/LR6) non-rechargeable lithium or alkaline batteries, or NiMH rechargeable batteries with 1.2–1.5 V nominal voltage.
- Observe correct polarity when inserting the battery.
- Remove the battery if the dosimeter will not be used for an extended period.
- Do not attempt to recharge non-rechargeable batteries.
- Dispose of used batteries in accordance with local regulations. Do not dispose of batteries with household waste.

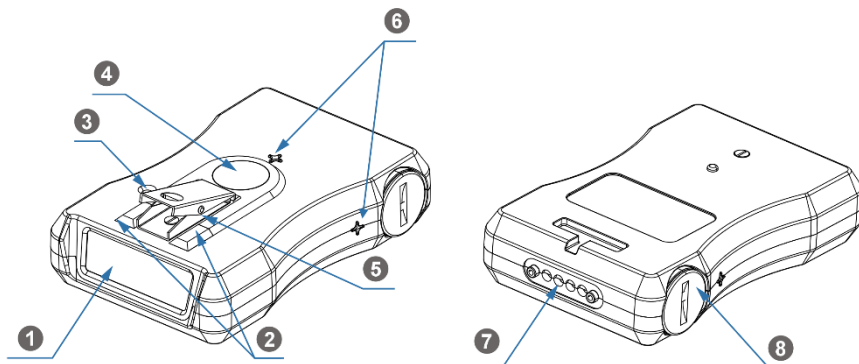
2.3 Tamper Protection

The dosimeter is sealed to prevent unauthorized access that could alter measurement results. Do not attempt to open or tamper with the sealed enclosure. The battery compartment is the only user-serviceable area.

3. Product Description

3.1 Components and Controls

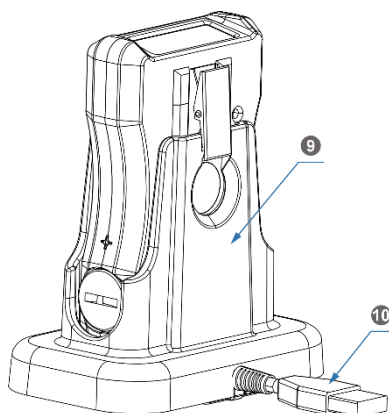
The MS24A dosimeter consists of the following external components (refer picture below):



Item	Description
1	LCD Display — Segmented LCD showing dose equivalent, dose equivalent rate, alarm status, and service indicators
2	Visual Alarm Indicator (LED) — Multi-color LED providing red, green and yellow alarm signals
3	Audible Alarm (Buzzer) — Piezoelectric sounder for dose and dose rate alarm tones
4	Control Button — Single push-button for mode switching, threshold viewing, alarm acknowledgment, and backlight activation
5	Pocket Clip — Reversible clip for pocket or outer-garment wearing configurations
6	Detector Center — Marked position of the radiation detector center for calibration geometry reference
7	Contact Pad — Hermetically sealed electrical contacts for USB/RS-485 data communication and battery charging via the docking station
8	Battery Compartment Cover — Screw-type cover with O-ring seal for IP67 water and dust protection

3.2 MS24A Reader

The MS24A Reader (docking station) serves as both a communication interface and a charging cradle (see the picture below):



It connects to a PC via USB and provides the following functions:

- Data transfer between the dosimeter and PC via USB
- Charging of NiMH rechargeable batteries when installed in the dosimeter
- Secure retention of the dosimeter during readout and configuration

The docking station (Item 9 on the device diagram) features a USB Type connector (Item 10 on the device diagram) for PC connection.

NOTE — When installing the MS24A into the Reader, make sure the clip is locked into the reader's holder. This provides secure retention of the dosimeter and reliable USB communication.

3.3 Detector Technology

The MS24A uses a Pb-doped polystyrene scintillator as its radiation detector. The lead (Pb) doping provides energy compensation that yields a response closely matching that of human tissue across the photon energy range of 40 keV to 7 MeV. This tissue-equivalent response eliminates the need for complex energy-dependent correction factors and ensures accurate dose assessment in mixed-energy radiation fields.

The scintillator converts photon radiation into light pulses, which are then converted to electrical signals by an integrated photodetector. The built-in microcontroller processes these pulses to calculate personal dose equivalent $H_p(10)$ and personal dose equivalent rate, applying statistical processing algorithms that provide fast response to changing radiation levels while maintaining measurement accuracy.

3.4 Marking and Sealing

Each dosimeter bears a self-adhesive label showing the manufacturer name or trademark, the model designation, the individual serial number (including year of manufacture), battery polarity diagram and the ingress protection rating (IP code). The dosimeter is sealed to prevent unauthorized access and to ensure the integrity of measurement results.

4. Technical Specifications

The specifications listed below define the performance characteristics of the MS24A EPD. All accuracy specifications apply under reference conditions as defined in IEC 61526:2024 unless otherwise noted.

Radiological Performance	
Detector type	Pb-doped polystyrene scintillator
Measured quantity, photon radiation	Personal dose equivalent Hp(10) and personal dose equivalent rate H*p(10)
Dose equivalent rate display range	0.01 μ Sv/h to 20 Sv/h
Dose equivalent rate measurement range	0.1 μ Sv/h to 10 Sv/h
Dose equivalent rate accuracy ($\pm$)	$\leq 20\%$ (reference conditions) within the measurement range
Dose equivalent display range	0.01 μ Sv to 100 Sv
Dose equivalent measurement range	1 μ Sv to 20 Sv
Dose equivalent accuracy ($\pm$)	$\leq 15\%$ within the measurement range
Energy range (photon)	40 keV to 7 MeV
Energy response (relative to Cs-137, 0.662 MeV) within the energy range	$\leq 30\%$

4.1 Angular Response (Anisotropy)

The following table shows the angular dependence of the dosimeter response relative to the calibration direction (0°) in horizontal and vertical planes, measured on an ISO slab phantom according to ISO 4037-3.

Angle (degrees)	Energy of photon radiation, MeV					
	Angular response, (δ_α , %)					
	Horizontal Plane			Vertical Plane		
	0.0595	0.662	1.25	0.0595	0.662	1.25
0	0	0	0	0	0	0
15	± 5	± 5	± 5	± 5	± 5	± 5
30	± 5	± 5	± 5	± 5	± 5	± 5
45	± 10	± 10	± 10	± 10	± 10	± 10
60	± 15	± 10	± 10	± 15	± 10	± 10
75	± 30	± 10	± 10	± 30	± 10	± 10
90	± 50	± 15	± 15	± 50	± 15	± 15
-15	± 5	± 5	± 5	± 5	± 5	± 5
-30	± 5	± 5	± 5	± 5	± 5	± 5
-45	± 15	± 10	± 10	± 10	± 10	± 10
-60	± 15	± 10	± 10	± 15	± 10	± 10
-75	± 50	± 15	± 10	± 50	± 10	± 15
-90	± 70	± 15	± 15	± 85	± 15	± 15

4.2 Environmental Influence on Accuracy

The maximum additional measurement uncertainty due to environmental factors:

- Temperature: $\leq 10\%$ additional uncertainty over the range -20°C to $+50^{\circ}\text{C}$ (relative to reference conditions at $20 \pm 5^{\circ}\text{C}$)

Alarm Threshold Settings

Dose equivalent alarm range	1 μSv to 20 Sv
Dose equivalent rate alarm range	1.0 $\mu\text{Sv/h}$ to 10.0 Sv/h
Dose equivalent and dose equivalent rate alarm step resolution	Any value within a measurement range. Configurable via software

Operational Characteristics

Display type	Segmented LCD with backlight
Display units	μSv , mSv, Sv (automatic ranging)
Data interfaces	USB or RS-485 (via MS24A Reader)
Non-volatile memory	$\geq 2,000$ events
Alarm types	Audible (buzzer), visual (LED), vibration
Two alarm thresholds per quantity	Dose equivalent and dose equivalent rate

Physical and Environmental

Dimensions (without clip)	23 × 63 × 91 mm
Weight (without clip and battery)	≤ 85 g
Power supply	1 × AA battery (lithium FR6, alkaline LR6, or NiMH rechargeable)
Battery life (continuous operation)	typically 3,000 hours (≈ 90 days) at normal operating conditions
Operating temperature	-20°C to $+50^{\circ}\text{C}$
Operating humidity	Up to 95% RH at 40°C
Atmospheric pressure	84 kPa to 106.7 kPa
Ingress protection (IEC 60529)	IP67
Drop resistance	1.5 m onto concrete floor

Reliability

Mean time between failures (MTBF)	$\geq 30,000$ hours
Mean time to repair (MTTR)	≤ 60 minutes

Service life	≥15 years
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Standards Compliance	
Personal dosimetry	ANSI N42.20, IEC 61526:2024
EMC immunity	IEC 61000-6-2 (industrial environments)
Ingress protection	IEC 60529 (IP67)
Environmental testing	IEC 60068 series
Quality management	ISO 9001:2015 (manufacturer certified)

5. Operating Instructions

5.1 Operating Modes

The MS24A has two operating modes:

Sleep Mode: A low-power standby state used during storage and maintenance. In this mode the display shows "OFF" and no measurements are performed. The dosimeter settings can be viewed and modified via the PC software while in sleep mode. Energy consumption is minimal.

Measurement Mode: The active operating mode in which the dosimeter continuously measures Hp(10) dose equivalent and dose equivalent rate, displays results on the LCD, monitors alarm thresholds, and records events to non-volatile memory.

5.2 Switching On (Sleep to Measurement Mode)

The dosimeter is shipped from the factory in sleep mode with "OFF" displayed on the LCD.

To activate the dosimeter:

1. Press the control button once. The "OFF" indication disappears.
2. The dosimeter performs a self-diagnostic cycle, verifying the detector, memory, and display functions.
3. Upon successful self-test, the dosimeter enters measurement mode and begins displaying the current dose equivalent or dose equivalent rate (configurable via software).

Alternatively, the dosimeter can be activated remotely using the Universal Configurator software via USB connection through the MS24A Reader.

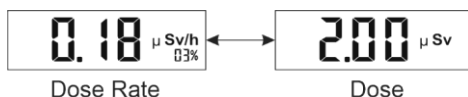
5.3 Display Screens

In measurement mode, the dosimeter provides two primary display screens:

Dose Equivalent Rate Screen: Shows the current H*p(10) dose equivalent rate in $\mu\text{Sv/h}$, mSv/h , or Sv/h with automatic unit scaling.

Accumulated Dose Equivalent Screen: Shows the accumulated Hp(10) dose equivalent in μSv , mSv , or Sv with automatic unit scaling.

To switch between the two screens, briefly press the control button.



The dosimeter returns to the default screen automatically after a timeout period.

The default screen can be selected via configuration software, see section 7.3 of this manual for details.

5.4 Display Symbols

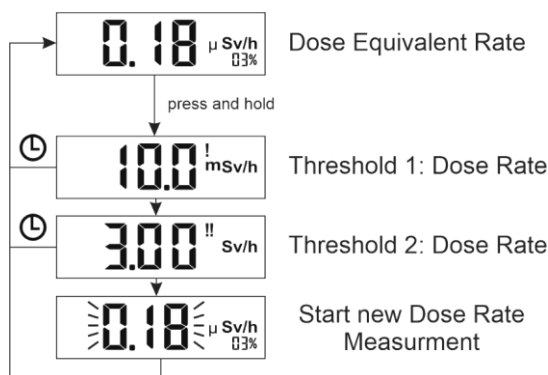
The following service symbols may appear on the LCD during measurement mode:

Symbol	Meaning
	Low battery indicator
!	First alarm threshold active (warning level)
!!	Second alarm threshold active (danger level)
XX%	Counting statistics quality indicator for dose rate measurement
	Vibration alarm is disabled
	Audible alarm is disabled
	Time display on screen

5.5 Viewing Alarm Thresholds

To view the currently programmed alarm threshold values during measurement mode:

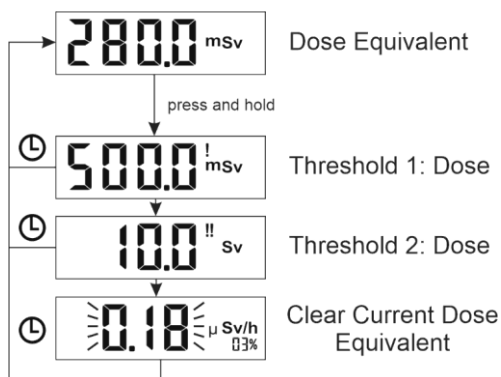
1. Press and hold the control button for 3 seconds on the dose equivalent rate screen.
2. The display cycles through: first dose rate threshold, second dose rate threshold, and (if enabled via configuration software) manual dose rate measurement initiation.



3. If no button is pressed within 2 seconds, the display automatically returns to the default screen.

Note that the last screen – Starting new Dose Rate Measurement – may be disabled via the configuration software. In this case only two alarm thresholds may be viewed by using the control button.

Similarly, pressing and holding the button on the accumulated dose equivalent screen displays the dose equivalent thresholds and (if enabled) the option to reset the accumulated dose.



Manual clearing of accumulated dose equivalent using the control button may be disabled via the configuration software; see section 7.3 of this manual for more details. In this case, only two alarm thresholds may be viewed by using the control button.

5.6 Backlight

The LCD backlight activates automatically each time the control button is pressed, providing visibility in low-light conditions. The backlight turns off after a configurable timeout to conserve battery life.

6. Alarms and Notifications

6.1 Alarm Threshold System

The MS24A provides a two-level alarm system for both dose equivalent and dose equivalent rate, following the conventions of IEC 61526:2024. Each quantity has a warning (first) and a danger (second) threshold level, configurable via the Universal Configurator software.

6.2 Alarm Descriptions

Alarm Event	Sound / Vibration	LED Indicator	Priority
Dose Warning (1st threshold)	1 long + 1 short beep/vibration every 2 seconds	Red, synchronized with sound	2
Dose Danger (2nd threshold)	1 long + 2 short beeps/vibrations every 2 seconds	Red, synchronized with sound	1
Dose Rate Warning (1st threshold)	1 short beep/vibration per second	Yellow, synchronized with sound	4
Dose Rate Danger (2nd threshold)	2 short beeps/vibrations per second	Yellow, synchronized with sound	3
Low Battery Critical	3 short beeps every 10 min (no vibration)	Red, 3 short flashes every 1 min	5

Lower priority number indicates higher alarm priority. When multiple alarm conditions exist simultaneously, the highest-priority alarm takes precedence for the audible and vibration indicators.

6.3 Alarm Acknowledgment

To acknowledge an active alarm and temporarily silence the audible and vibration signals, press and hold the control button for more than 3 seconds. The visual (LED) alarm remains active even after acknowledgment.

For dose rate alarms, the visual indicator remains active until the dose equivalent rate drops below the threshold level. Dose equivalent alarms persist because the accumulated dose cannot decrease during operation.

6.4 Low Battery Indication

When the battery voltage drops below the critical level, a low battery indicator appears in the lower-right corner of the display. The dosimeter provides audible and visual alerts as described in the alarm table above. Replace the battery promptly to avoid interruption of dose monitoring. The dosimeter retains all accumulated dose data and event history in non-volatile memory during battery replacement.

7. Configuration Software

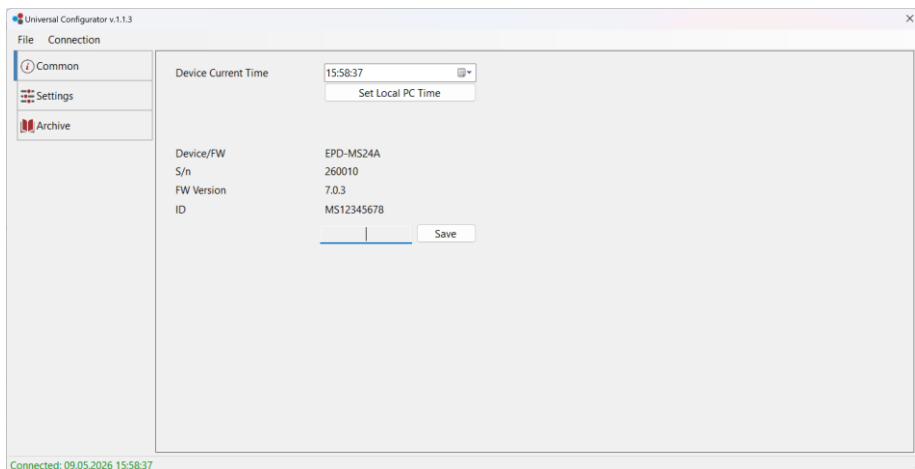
7.1 Universal Configurator

The MS24A settings are managed using the Universal Configurator PC software supplied by Micasensor. This software allows authorized personnel to read, modify, and write dosimeter parameters, as well as retrieve stored measurement history.

7.2 Connecting to the PC

1. Insert the docking station USB cable into an available USB port on the PC.
2. Place the dosimeter into the docking station, ensuring proper alignment with the contact pads.
3. Launch the Universal Configurator software. For the USB interface version of the MS24A, the instrument connects automatically and no proprietary driver installation is needed.
4. For the RS-485 interface option of the MS24A (used mainly for group storage and operation of dosimeters by the dose management system), install the system drivers first (drivers' folder), then select the appropriate COM port from the port selection list. Click the Connect button to establish communication.

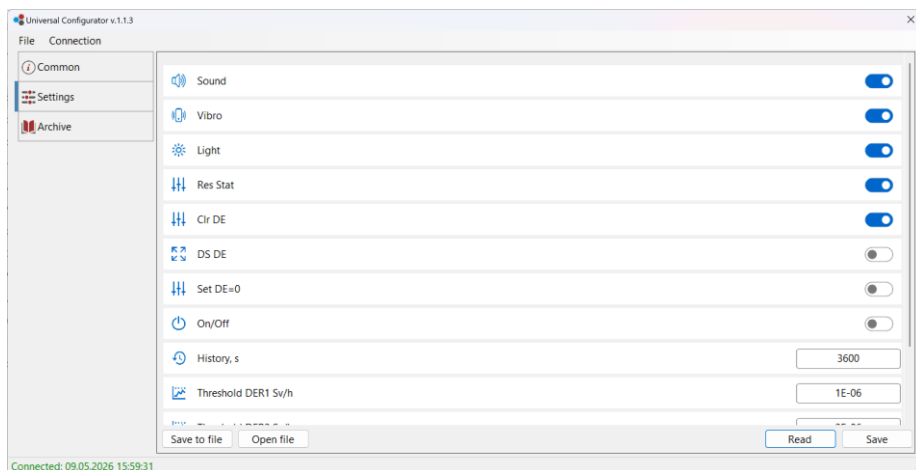
When connection is established, the following screen appears (see below picture):



On the first connection and after replacing the battery, press “Set Local PC Time” to synchronize the MS24A clock with the PC time. This is an important step, as all events recorded in the non-volatile device memory are timestamped with the local instrument time.

7.3 Available Settings

On the “Settings” tab, the Configurator provides access to the following parameters (see picture below):



Below is detailed information about available settings:

Parameter	Description
 Sound	Audible alarm enable/disable
 Vibro	Vibration alarm enable/disable
 Light	Light alarm enable/disable
 Res. Stat.	Enable/disable manual start of new measurement by User (by control button)
 Clr. DE	Enable/disable manual clearance of accumulated dose equivalent by User (by control button)
 DS DE	Set the dose equivalent the default screen
 Set DE=0	Clear accumulated Dose Equivalent
 On/Off	Switch the MS24A into Sleep mode (OFF message on the screen) and back
 History, s	Set the interval of logging measurement data into the device memory, in seconds
 Threshold DER1 Sv/h  Threshold DER2 Sv/h	Set the dose equivalent rate alarm thresholds (warning and danger levels), in Sv/h
 Threshold DE1 Sv  Threshold DE2 Sv	Set the dose equivalent alarm thresholds (warning and danger levels), in Sv

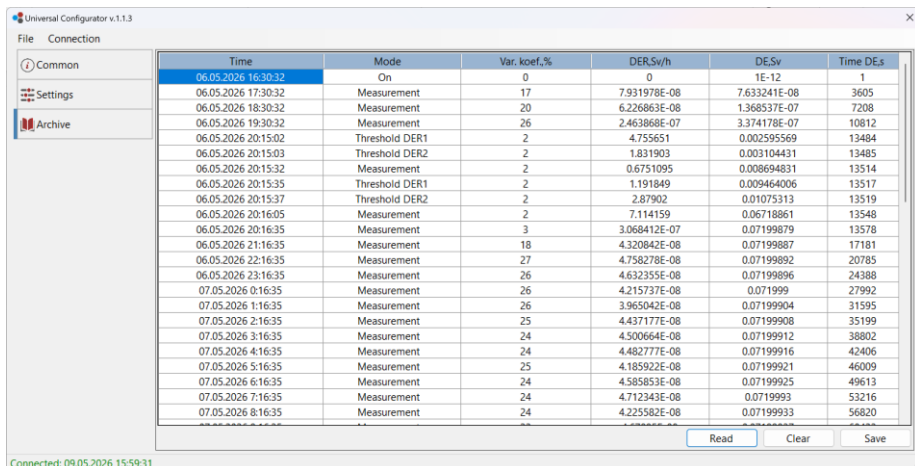
7.4 Reading and Writing Settings

To read current settings, navigate to the Settings tab in the Configurator and click Read. To modify a setting, enter the new value in the corresponding field and click Write. The new values are immediately stored in the dosimeter non-volatile memory.

For the MS24A with the USB interface, reading settings is performed automatically. However, to apply new settings, press the Write button.

7.5 Reading Data Archive

To read the measurement data logged into the device's memory, navigate to the Archive tab in the Configurator and click Read (see below picture):



The measurement data are logged into the device memory with the time interval set on the Settings tab. The recording mode is cyclic, meaning that after all available memory is filled up, the older records will be automatically replaced with the new ones. Keep this in mind when selecting the history recording interval.

To export data into the Excel format and save it locally on PC press the Save button. To clear all data from the device memory, select Clear.

For the MS24A with the USB interface, reading data from the device memory is performed automatically.

8. Maintenance

8.1 Routine Maintenance

The MS24A requires minimal routine maintenance. Periodic inspection and cleaning are recommended:

- Visually inspect the dosimeter for physical damage, cracks, or seal integrity.
- Clean the exterior with a soft, dry cloth. If decontamination is required, wipe with a soft cloth dampened with ethyl alcohol. Do not use abrasive cleaners or solvents.
- Verify that the contact pads are clean and free of corrosion.
- Check the battery compartment O-ring seal for damage or deterioration during battery replacement.

8.2 Decontamination

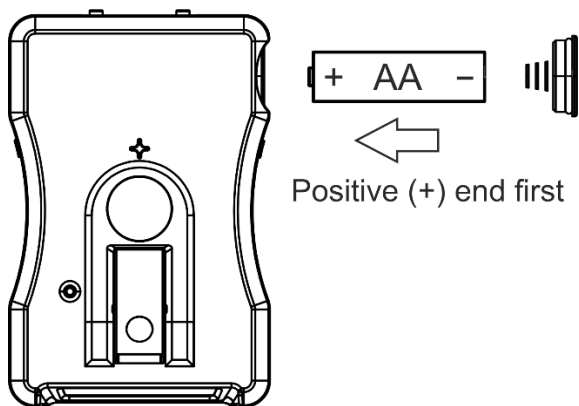
If radioactive contamination is detected on the dosimeter housing, decontaminate by gently wiping the exterior with a soft cloth moistened with ethyl alcohol. The smooth housing surface is designed for easy decontamination. If contamination persists after cleaning, contact Micasensor or your radiation safety officer.

9. Battery Replacement

Replace the battery when the low-battery indicator appears on the display or when battery voltage is insufficient for reliable operation. Note that all accumulated dose data and event history are preserved in non-volatile memory during battery replacement.

To replace the battery:

1. Using a flathead screwdriver or coin, unscrew the battery compartment cover by turning counterclockwise.
2. Remove the depleted battery.
3. Insert a new AA battery, observing correct polarity (positive end first) as indicated on the picture below:



4. Replace the battery compartment cover, turning clockwise. Ensure the rubber O-ring seal is properly seated to maintain IP67 water and dust protection.

CAUTION — Verify the O-ring is undamaged and correctly positioned before closing the battery compartment. A damaged or displaced O-ring may compromise the IP67 ingress protection rating.

Compatible battery types:

- Lithium primary: FR6 (FR14G505) 1.5 V (recommended for longest battery life)
- Alkaline primary: LR6 (AA) 1.5 V
- NiMH rechargeable: AA size, 1.2 V nominal (chargeable via docking station)

10. Reversing the Pocket Clip

By default, the clip is installed by the manufacturer facing the direction of irradiation (opposite to the body side). This allows the instrument to be kept inside a chest pocket, secured by the clip to the pocket edge. This is the recommended way to wear the MS24A EPD, as it ensures the instrument will not fall or be accidentally damaged even if the clip detaches.

However, there is another way to fix the pocket clip for the scenario of wearing the instrument on the outside side of the pocket. In this case the clip shall be fixed on reversed side of the instrument. To do this, unscrew the screw fixing the clip, and place it on another side the same way – fixing by the screw.

11. Calibration and Verification

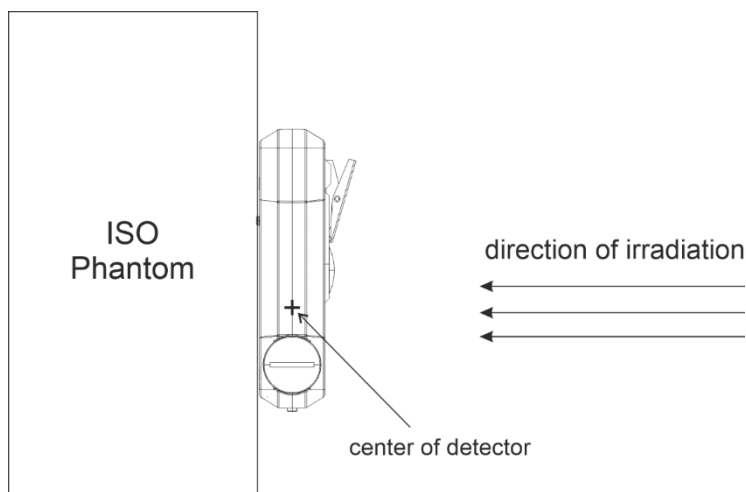
11.1 Periodic Verification

Periodic verification (calibration check) of the MS24A is recommended to be performed once per year, in accordance with the verification procedures accepted in your facility and approved by Micasensor. Verification ensures that the dosimeter continues to meet its stated accuracy specifications.

Calibration and verification must be performed by qualified personnel using traceable reference radiation sources at an accredited calibration laboratory in compliance with the requirements of IEC 61526:2024 and ANSI N42.20.

11.2 Calibration Geometry

When positioning the dosimeter for calibration, align the radiation beam with the detector center mark (Item 6) on the dosimeter housing. The dosimeter should be mounted on an ISO water slab phantom or appropriate phantom per IEC 61526:2024 requirements for Hp(10) calibration. The direction of irradiation is provided on the picture below:



11.3 Recording Results

Document all verification and calibration results according to your facility's quality assurance procedures and the dosimeter passport. Affix verification labels or certificates as required by local metrology regulations.

12. Storage and Transportation

12.1 Storage Conditions

In original packaging: Temperature -40°C to $+50^{\circ}\text{C}$, relative humidity up to 95% at 40°C . Maximum storage duration shall not exceed the rated service life.

Without packaging: Temperature $+10^{\circ}\text{C}$ to $+35^{\circ}\text{C}$, relative humidity up to 80% at 25°C . The storage area must be free of dust, acid or alkali vapors, aggressive gases, or other corrosive agents.

Remove batteries before long-term storage to prevent leakage and corrosion.

12.2 Transportation

The MS24A in its original packaging may be transported by any enclosed means of transport under the following conditions:

- Temperature: -40°C to $+50^{\circ}\text{C}$
- Relative humidity: up to 95% at 40°C
- For maritime transport: place packaged dosimeters in sealed polyethylene bags with silica gel desiccant.
- For air transport: store in pressurized (climate-controlled) cargo compartments.

13. Disposal

At end of service life, the MS24A dosimeter should be disposed of in accordance with local regulations for electronic waste (WEEE or equivalent).

Before disposal:

1. Remove the battery from the dosimeter.
2. Dispose of the battery separately in accordance with local battery disposal regulations. Do not dispose of batteries with household waste.
3. If the dosimeter has been used in potentially contaminating environments, survey for radioactive contamination before disposal. Contact your radiation safety officer if contamination is detected.

14. Troubleshooting

The following table provides guidance for common operational issues:

Symptom	Possible Cause	Corrective Action
Display shows "OFF"	The instrument is in Sleep mode	Press the control button to switch the instrument ON
Display is dim or partially visible	Low battery or extreme temperature	Replace battery; allow dosimeter to acclimate to operating temperature range
No audible alarm when threshold is exceeded	Audible alarm disabled in Configurator	Enable audible alarm via Universal Configurator software
No vibration alarm	Vibration alarm disabled in Configurator	Enable vibration alarm via Universal Configurator software
Dosimeter not recognized by Configurator software	Incorrect COM port selected; dirty contacts; faulty USB cable	Select correct COM port; clean contact pads; try a different USB cable
Dose rate reading appears abnormally high	Strong radiation field or electromagnetic interference	Move to a lower-radiation area; verify reading with a second instrument
Battery life shorter than expected	High alarm activity; extreme temperatures; aged battery	Reduce alarm frequency if possible; store within temperature range; use lithium FR6 battery for longest life
Self-test failure on startup	Internal fault detected	Contact Micasensor service or authorized maintenance facility

15. Ordering Information

The standard delivery kit of MS24A includes:

Item	Q-ty
MS24A Electronic Personal Dosimeter	1
AA-type battery (alkaline)	1
User Manual and Universal Configurator Software supplied on electronic media (USB flash-drive)	1
Calibration certificate	1
Individual package (carton box)	1

The following items are also available for order. Contact Micasensor for pricing and delivery:

Item	Note
MS24A Reader	USB interface, includes charging capability in case of rechargeable battery
User Manual (printed)	Usually, 1 copy per 50 dosimeters standard
Group pelican case package	Usually, 1 per 10 dosimeters. Custom pelican cases for different quantities of dosimeters are available

16. Warranty and Liability

16.1 Warranty

CBRN PROTECTION TCT BV warrants to the purchaser (the “Purchaser”) that MS24A Electronic Personal Dosimeter (the Product), including component parts, to be free from defects in material and workmanship, under normal use and service for a period of one year (the “Warranty Period”) provided, however, that the foregoing warranties are expressly contingent (and shall otherwise be void) upon use of the Product in accordance with specifications, this service manual and without misuse, abuse, or abnormal use, accident, damage, alteration, or modification thereto or improper or unauthorized repairs and maintenance.

Non-substantial variations of performance from the documentation do not establish a warranty right.

16.2 Liability Disclaimer

CBRN PROTECTION TCT BV. endeavors to assure that its published engineering specifications and manuals are correct. Despite these efforts, errors can occur. CBRN PROTECTION TCT BV. reserves the right to correct any such errors and disclaims all liability resulting therefrom.

16.3 No Liability for Consequential Damage

In no event shall CBRN PROTECTION TCT BV., any of its affiliates, or anyone else involved in the creation, production, delivery of the accompanying product (including hardware and software) be liable for any loss of profits or any special, incidental, consequential, exemplary, or other damages whatsoever (including without limitation, damages resulting from cost of substitute procurement, loss of use, loss of data, loss of savings, loss of revenue, loss of business or failure or delay in performance) arising out of the use of or the results of use of or inability to use such product, even if CBRN PROTECTION TCT BV. and its affiliates have been advised of the possibility of such damages.

